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Prospects of Multi-Object Spectroscopy in advanced Amateur Astronomy

Semester Project

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Abstract

Context: Multi-object spectroscopy is one of the backbones of modern astronomy. While available at most large telescopes, small telescopes have no options available at the moment to perform MOS.

Aim: A concept for a spectrometer using a Digital Mirror Device as a variable slit mask for small telescopes is evaluated.

Methods: Zemax was used to evaluate the optical performance of the proposed device, simulated stellar spectra were used to evaluate scientific use-cases.

Results: The proposed device has a field of view of 14 by 9 arcminutes, a spectral resolution $R \approx 200$ and a throughput of 19%. It can resolve strong spectral features in stellar spectra and allows us to distinguish the phases of variable stars.

Conclusion: The device can only be used for regions with a high star density and stars up to magnitude 13. This limits its scientific value. In the teaching setting on the other hand, the device can provide hands-on experience with MOS devices at a relatively low cost.

Contents

Contents	ii
1 Introduction	1
2 Science case	5
3 Proposed optic setup	7
4 Expected performance	10
5 Discussion	15
Bibliography	18
List of Figures	22
6 Acknowledgement	25
A Appendix	27
A.1 Zemax configuration	27
A.2 Code	27
A.3 DMD datasheet	27
A.4 Telescope datasheet	27
A.5 Simulated spot diagrams	28

Chapter 1

Introduction

Multi-object spectroscopy (MOS) has revolutionized observational astronomy, enabling simultaneous spectral analysis of multiple objects within a single field of view. While this technology has become the standard in large professional observatories, the application on small telescopes remains largely unexplored. This thesis investigates the prospects of implementing a cost-effective multi-object spectrometer on a small observatory, opening new possibilities for educational programs and advanced amateurs in observational astronomy.

State of the art MOS devices are used for both stellar observation campaigns and galactic studies. The large multiplexing allows collecting statistically significant number of observations of a target class to test current theory on the evolution of the Milky Way, galactic evolution and first galaxies [1].

Most multi-object Spectroscopy setups follow a similar schematic. Figure 1.1 shows the different conceptual elements of a MOS system. In this paragraph different implementations of the target selection are introduced and an overview of the current state of the art in the professional world of MOS systems is given.

The most basic spectrometer uses a slitless setup. A grating, prism or grism is inserted in a pupil plane or collimated beam, the incoming light split up into its wavelengths and refocused on a detector. With the WFPC1 and later the ACS instrument [2] the Hubble Space Telescope was equipped with slitless spectroscopy capabilities from its launch. Both NIRCam [3] and NIRISS [4] on JWST are equipped with slitless spectroscopy capabilities. The implementation of a grism is relatively easy, as it can be mounted in a slot of the filter wheel of the imaging instrument. The main downside of slitless spectroscopy is the overlap of spectra from different sources, contamination from the background, as well as reduced precision from extended sources.

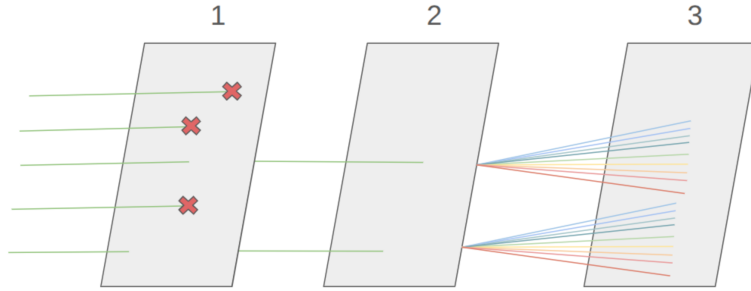


Figure 1.1: The goal of multi-object spectroscopy is to capture spectra of multiple objects in one observation. Most MOS systems follow the same scheme. In a focal plane (1) targets are selected, their light dispersed (2) and the spectra captured (3).

Additionally, the mapping from retrieved spectra onto a wavelength scale is not trivial.

One way to deal with these issues is by placing a slit over an object that is of special interest. The light from nearby sources is thereby excluded. For spatially extended sources, a slice can be selected through careful positioning of the slit. The signal-to-noise ratio increases, as the background contamination now only originates from the slit, instead of being an overlay of background spectra from all positions. While these advantages result in improved measurements, the observation time at large Observatories is too scarce to use on a single target if other options are available.

By manufacturing a plate with slits just in the right place, multiple targets can be selected at once. The position of the targets is unconstrained, but too closely spaced slits still result in overlapping spectra. The main issue with this approach is that each target selection needs a custom slit plate. The W. M. Keck observatory has multiple instruments using this setup. DEIMOS [5] and LRIS [6] both use a slit mask milled out of thin aluminum sheet. An issue that can emerge here is the alignment of the plates, which can lead to a significant amount of lost observation time [7]. MOSFIRE [8], shown in figure 1.2, on the other hand uses a remote controlled configurable slit unit, allowing for changes to the configuration on the fly.

A related approach is a micro shutter array. Instead of slits in an aperture, a plate with trapdoors that can be opened independently is used to select targets. NIRSPEC on JWST uses this technique to observe up to 100 targets at once [9]. One of the main disadvantages is that unwanted targets are simply blocked. If multiple targets are close by along the axis of dispersion, the spectra will still overlap on the detector. This limits the achievable target density.

An approach to solve target spectra overlap is to use optical fibers that are placed in a focus plane. In early iterations of this technique, the fibers

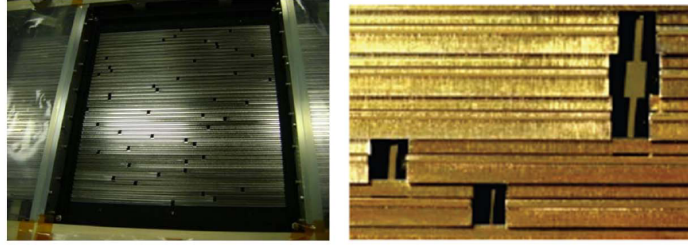


Figure 1.2: Images of the robotically controlled slit plate of the MOSFIRE instrument at the Keck Observatory. The image was sourced from [10].

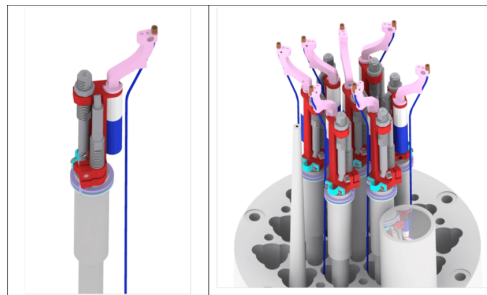


Figure 1.3: The Fiber Positioning Unit (FPU) of the Moons instrument allows to precisely position fibers on targets. As the reach of the FPUs overlaps, multiple spectra of locations close by can be obtained. The figure was sourced from the original Moons Paper [1].

were placed manually in position. For the 92 fibers of FLAIR this process could take 4 to 6 hours [11]. To speed up the process and thereby increase observation efficiency, robotic pick and place systems were developed. The 2df instrument on the Anglo-Australian Telescope consists of two field plates. While one is used for observation, the other one can be automatically reset, resulting in no lost observation time for integrations longer than 45 min [12]. A further speedup in repositioning can be achieved by using robotic arms directly that hold the fibers in place, as they can move in parallel. Different mechanical designs for the arms have been developed. Figure 1.3 shows the fiber positioning unit of the Moons instrument. It uses a combination of two parallel rotating joints, where one is off centric to select targets [1]. The 4MOST instrument mounted on the VISTA telescope uses a pivoting arm [13]. Both approaches allow fibers to be placed close to each other. As the translation of each arm is limited, the number of fibers that can be close by is limited. One approach to solving this is starbugs [14]. Small concentric tubes are attached to a glass plate using a vacuum, piezoelectric “legs” are used to move the tubes. The tubes can be moved independently over the whole glass plane. While the minimal distance is larger than with robotic arms, as the fibers are located in the middle of the inner tube, more fibers can be positioned in the same area.

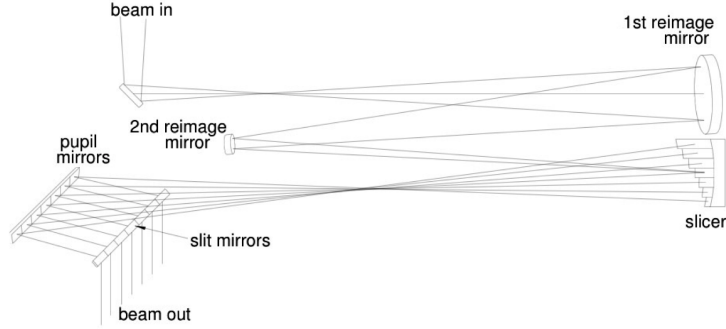


Figure 1.4: Schematics of the integral field spectrometer of the NIRSpec instrument on JWST. The figure was sourced from [18].

For dense targets like galaxies, a higher spatial sampling rate is needed. Integral field units (IFU) can provide this using different techniques. One common way is to use fiber bundles. These fiber bundles can be either fixed in place, or positioned with one of the above-mentioned methods. The WEAVE instrument implements single fibers and IFU fiber bundles that can be automatically positioned [15]. Another approach is image slicers. They use mirrors to separate the incoming light before sending it through a spectrometer. MUSE at the VLT first splits the field of view into 12 and slices each of them independently, achieving a field of view of $1'$ by $1'$ [16]. NIRSpec [9] and Miri[17] on JWST both deploy an IFU spectrometer using a slicer. Figure 1.4 shows the schematics of the slicer used in NIRSpec.

The approach we will analyze further in this report is to select targets using a digital micro-mirror device (DMD). The basic working principle is similar to slit mask based systems. In the DMD case, the slit mask comprises individually controllable micro mirrors with sizes down to $5\ \mu m$. This approach has been explored by several concept studies [19], [20], [21] have been conducted, but no device has been deployed on a large telescope. DMDs are manufactured on a large scale for use in consumer electronics such as projectors and are widely available. The solutions presented here focus on a low-cost implementation at a small telescope.

Chapter 2

Science case

When developing possible science cases for such a potential spectrometer we have to consider some limiting outside factors. The MOS device would be mounted on the telescope of the ETHz physics department, a CDK20 from planewave. It has an aperture $f/6.8$ and a focal length of 3454 mm. In combination with a DMD with the physical size of 13.8 mm by 8.6 mm that translates to a field of view of around 14 by 9 arc-minutes on the night sky. Possible targets should have multiple objects of interest in this area. There are different structures of that size on the night sky. Venus, Jupiter and Saturn are all below one arc-minute. In the case of Jupiter there are structures visible that can be resolved considering the resolving power of 0.5 arc-seconds of the telescope. This resolution power translates to $4\text{ }\mu\text{m}$ on the DMD and fits well together with the mirror pitch of $7.6\text{ }\mu\text{m}$. Another set of interesting extended structures are globular clusters. They are densely packed groups of stars found around galaxies. We want to investigate possible use-cases for the MOS device using such a globular cluster as an example.

M5 is such a globular cluster. With a spatial extent of 23 arcminutes it is larger than the available field of view, but its bright core does contain a lot of stars with magnitude lower than 13 [22] that can be resolved in the spectrometer. It lays in the constellation serpens and contains around 100'000 stars.

More than 100 of these stars are variable[23]. More than 80 of them are RR-Lyrae, a special type of variable stars. As for the Cepheids, there exists a relationship between the period of the variability and the mass of the star for RR Lyrae stars.

Typical use-cases of spectroscopy in astronomy are the measurement of metallicity, radial velocity, identification of different stellar populations, iden-



Figure 2.1: The globular cluster Messier 5 observed with a 6 in f/9 amateur telescope. The image was sourced from <https://app.astrobin.com/equipment/explorer/telescope/1097/gso-6-f9-ritchey-chretienon25.08.2024>.

tification of main sequence stars and giant branch stars.

A possible measurement campaign could first measure as many bright stars as possible and categorize them. If any variable stars are found in this survey, they can be targeted in a follow up. Non variable stars can be used as a reference to reduce errors in the spectra of the target variable stars. Such errors could be clouds during parts of the observation or changing background light. As some lighting in the urban environment, such as building illumination and parts of the streetlight, are turned off at some point in the night, having a reference in the same frame allows one to correct for such effects or reject an observation if needed.

RR Lyrae are varying stars with a relatively short period [24]. Their surface temperature and radius change significantly during this period. These changes can be measured in the spectra of stars. Especially the temperature has an effect on the line spectrum and the lineshape. As the period of RR Lyrae stars is relatively short, multiple spectra can be taken over one night. From these spectra, the period and the metallicity can be retrieved, enabling one to calculate the distance to the cluster using a relation between the period and the metallicity.

Chapter 3

Proposed optic setup

Digital Mirror Devices are micro-electromechanical devices that contain an array of small mirrors that can individually be switched between an on and an off state, each tilted to one side. Figure 3.1 shows a schematic of a single mirror as well as an image of the mirror array. When using them in a multi object spectrometer, one side is attached to a spectrometer and captures the spectrum of the targets, the other side is attached to an imaging system for target selection and guiding during long observations. Figure 3.2 shows the concept on how targets can be selected. For this case study we evaluate the performance of a Texas Instruments DLP651NE, as chips like this are commonly found in home projectors. It has a resolution of 1920 by 1080 micro-mirrors with a pitch of $7.6\text{ }\mu\text{m}$.

The spectrometer design is shown in figure 3.3. All parts of the device lie in one plane. A caveat is that the collimator lens is tilted to compensate for the field curvature introduced by the DMD. This must be taken into account in the alignment process. As all parts but the dmd measure 50 mm in diameter, respectively 50 mm squared for the grating, the requirements for

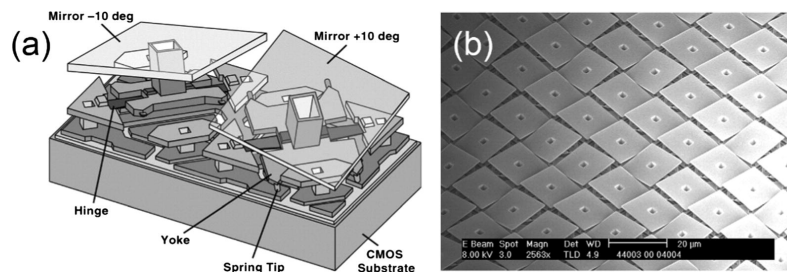


Figure 3.1: Schematics of the micro-mirror array of an DMD. Each micro-mirror can be individually set. The switching between the two stable states can happen at a high frequency. The image was sourced from <https://www.sciencedirect.com/topics/engineering/digital-micromirror-device>

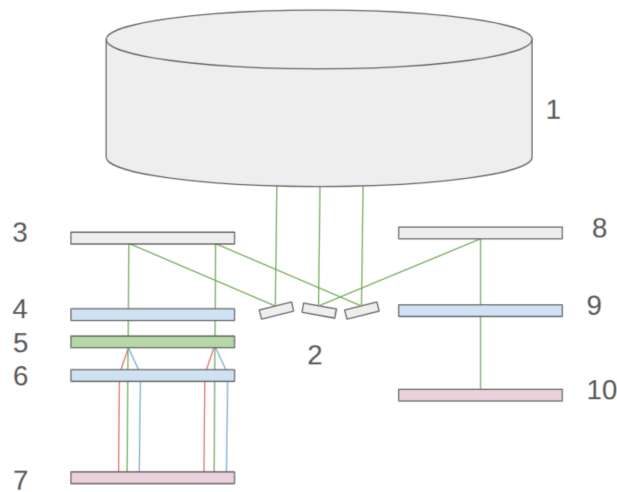


Figure 3.2: Schematic representation of the opto-mechanic setup using separate optical arms for spectroscopy and imaging. The light from the telescope (1) is split by the DMD (2) into the Spectroscopy setup (left arm) and the imaging setup (right arm). The light in the spectroscopy setup is redirected by a folding mirror (3) onto a collimator lens (4) passes through a grating (5), is refocused by the focusing optics (6) onto the spectroscopy sensor (7). The light in the imaging arm is redirected by a folding mirror (8) and is focused by an imaging lens (9) onto the imaging sensor (10).

the enclosure are relatively simple.

The imaging arm is needed for target selection and guiding during long exposures. It uses standard photography components to focus the light onto a second camera. As seen before with the spectrometer arm, we also face the issue of the tilted DMD that is imaged. This problem can be solved by using a macro tilt-shift lens. These lenses are readily available for different camera systems. Depending on the exact lens used, a folding mirror has to be used to have enough space on the back of the telescope. As photography lens designers do not share exact information about their lenses, no precise measure of the performance of these lenses can be given.

The cost of the device is relatively small, as only off the shelf components are used. The DMD chip can be extracted from a decommissioned projector. In the university environment projectors are replaced on regular maintenance intervals and often contain a fully functioning DMD chip. The cost of all standard parts is listed in table 3.1. This includes all optical components, as well as their mounts. The cost of the housing is not factored in, as the cost depends heavily on whether it can be manufactured in house or has to be hired out.

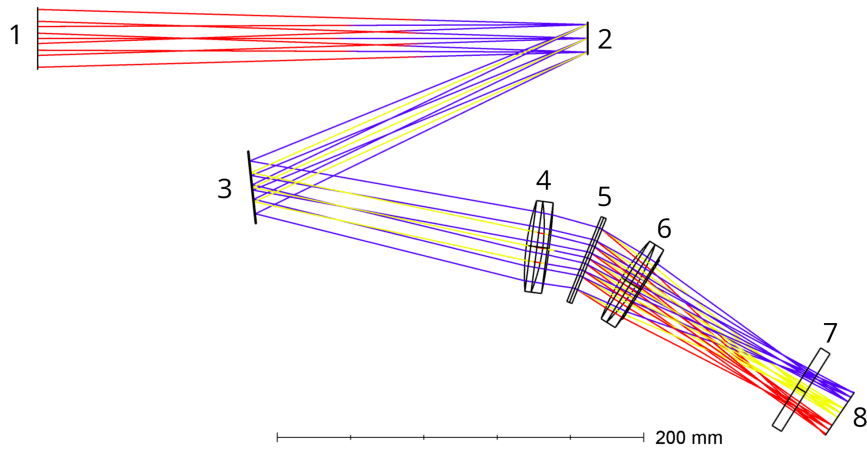


Figure 3.3: Optical layout of the spectroscopic arm. Light from the telescope (1) is modulated by the DMD (2) at the focus of the telescope, redirected via a folding mirror (3) to accommodate the telescope mounting geometry, collimated (4), dispersed by a 600 lines/mm transmission grating (5), focused (6), chromatically corrected by a flat window (7), and detected at the sensor (8). Both lenses have a focal length of 200 mm. The optical system was designed using Ansys Zemax OpticStudio. Rays of wavelength 700 nm (red), 580 nm (yellow), 460 nm (blue), from fields with an angle $+0.125^\circ$, 0° , -0.125° were traced.

Function	Edmund Optics Part Number	Amount	Cost Per Part
Folding Mirror	45-608	2	113.00 €
Collimator Lens	49-286	1	192.00 €
Grating	49-584	1	236.00 €
Focusing Lens	49-286	1	192.00 €
Window	37-022	1	135.00 €
Round Mounts	15-867	5	92.00 €
Square Mount	54-995	1	122.00 €
Tilt Shift Lens		1	
Total			1,563.00 €

Table 3.1: Cost of the proposed device excluding the cost for the enclosure and the imaging arm. All prices were retrieved 01.10.2025

Chapter 4

Expected performance

The proposed setup has spectral resolution of more than $R \approx 206$ over the wavelength range 260 nm - 700 nm. The transmission throughput of the device is 19% in the worst case. Both spectral resolution and throughput are better in the central wavelength range. To assess what these numbers mean for an actual measurement we calculate an observation of a Globular cluster using a planewave CDK20 in Zürich to observe the globular Cluster M5 from before.

To have full control over the parameters, we use simulated stellar spectra[25]. The complete grid of simulated spectra is available on the University of Göttingen website [26]. For performance analysis a variable star RR Lyrae with a surface gravity $\log(g)$ cycling between 2.5 dex and 3 dex and an effective temperature cycling between 6200 K and 6700 K[24]. The star is assumed to be of magnitude 11. The magnitude of the spectrum was kept constant. Figure 4.1 shows such a high resolution simulated spectrum.

In an urban location light pollution is a significant factor. As data for background spectra around Zürich are not available, we used a study from Vienna[27] to obtain the shape of a typical urban background spectrum and constructed our own using a spectral library[28] containing different lamp spectra. Figure 4.2 shows the spectrum of different lamp types. The assumption that artificial light from lamps dominates the background spectrum is justified in regard to the World Atlas of Artificial Sky Brightness[29] that estimates the ratio of natural to artificial light in the Zürich metropolitan area is more than 1 to 10. From the World Atlas of Artificial Sky Brightness we can also deduce that the brightness of the sky glow is between 16 and 19 mag sr⁻². In Zürich, more than 51% of streetlights are LEDs[30]. All of this information can be used to generate a probably nighttime background spectrum. A linear combination of half LED and half sodium lamps is used

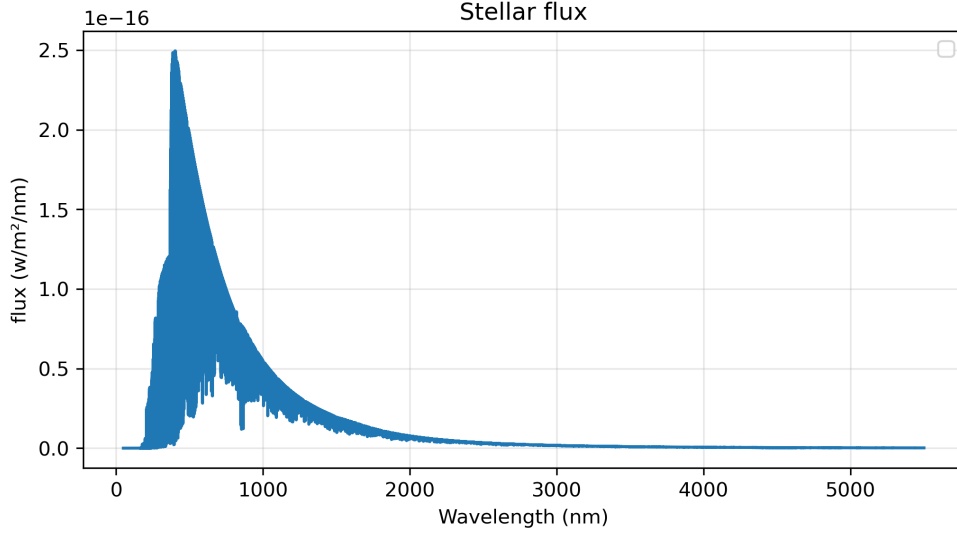


Figure 4.1: Spectrum of a 12 mag star simulated with phoenix[26]. The simulated star has an effective temperature of 6200 K, surface gravity $\log(g)=2.5$, $[\text{Fe}/\text{H}]=0$, $[\alpha/\text{Fe}]=0$, corresponding to a RR Lyrae star from [24].

and scaled to 16 mag. The assumed background spectrum is shown in figure 4.3.

Considering these inputs we can calculate the expected results of a 60 min exposure. Figure 4.4 shows the normalized spectrum that can be measured.

We can now analyze how the spectra of the two extrema of the cycle compare. To achieve this, we divide the spectra from different phases by each other to get the relative difference between the two. Figure 4.5 shows the ratio of the spectra in different phases.

So far we only showed that the spectral resolution of the device is high enough to resolve spectral differences between the different phases of an RR-Lyrae star. We must take the noise floor into account. It has to be low enough to retrieve the differences between the phases. The SNR can be calculated using formula 4.1, taken from [31]. P denotes the input flux, B the background flux, D the dark current, N_r the readout noise, Q_e the quantum efficiency of the sensor, t the integration time and $\Delta\lambda_p$ the wavelength range that is mapped to a single pixel.

$$SNR = \frac{S}{N_{tot}} = \frac{P Q_e t n_b / \Delta\lambda_p}{\sqrt{((P Q_e + B Q_e) / \Delta\lambda_p + D) t n_b + N_r^2}} \quad (4.1)$$

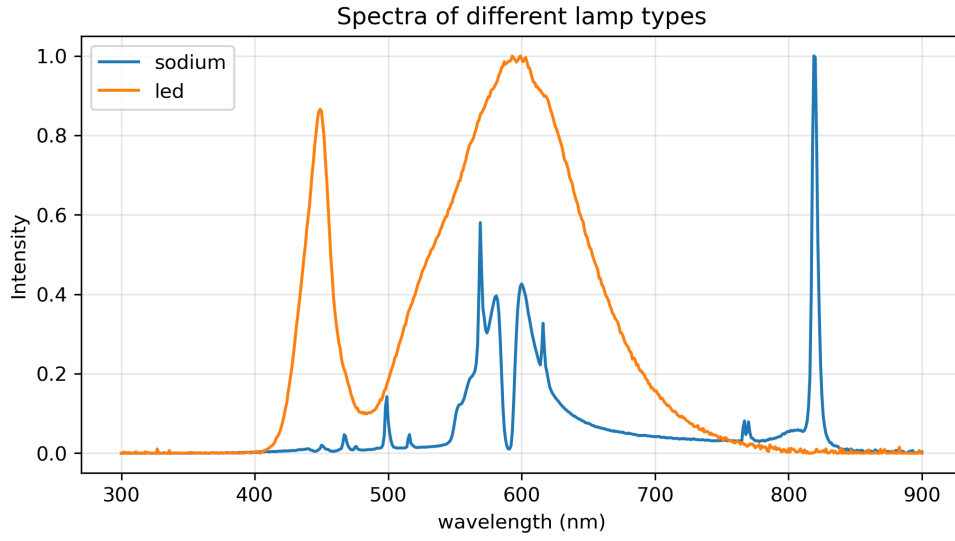


Figure 4.2: Spectra of different lamps. Samples of street light from a spectral library[28] are used.

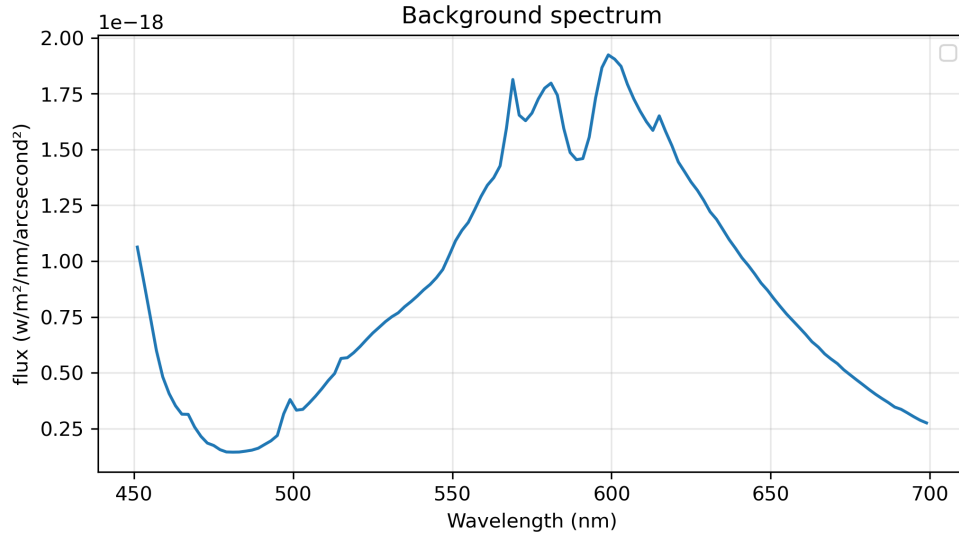


Figure 4.3: Proposed background spectrum at a 16 mag/arcsec²

We can compute the spectrally resolved SNR, shown in figure 4.6. The SNR is over 40 throughout the spectral range. This translates to an error of 2.5%. The spectral features in figure 4.5 are about 5% over the average and therefore should show in the measurement.

The spectrum shows two clear absorption lines at 486 nm and 656 nm. Both of these lines are hydrogen lines. All other absorption features are below the

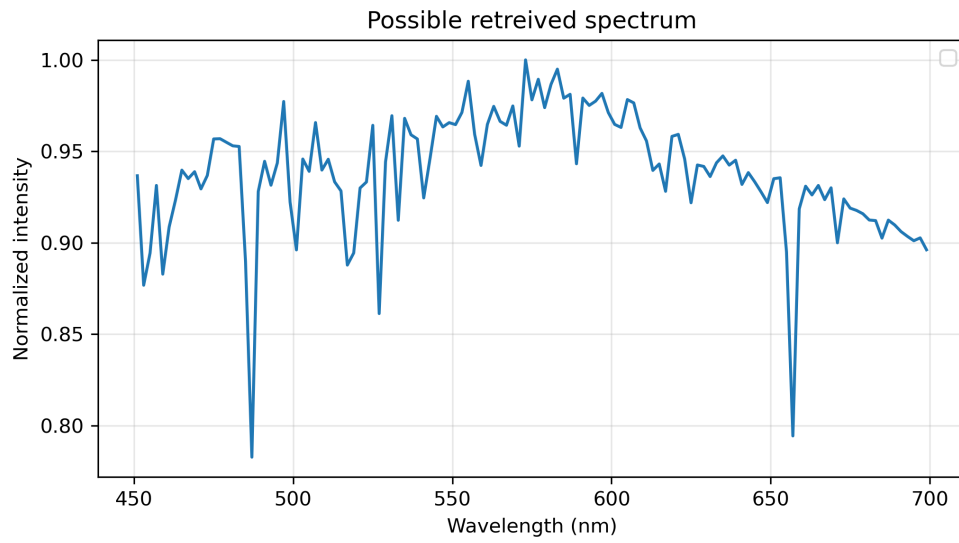


Figure 4.4: Possible retrieved spectrum from the mag 11 star using the proposed device. Two strong absorption features are clearly identifiable. The background and dark images were subtracted and the image normalized so that the highest value is 1.

noise floor. Metallicity computations are not possible with the given setup.

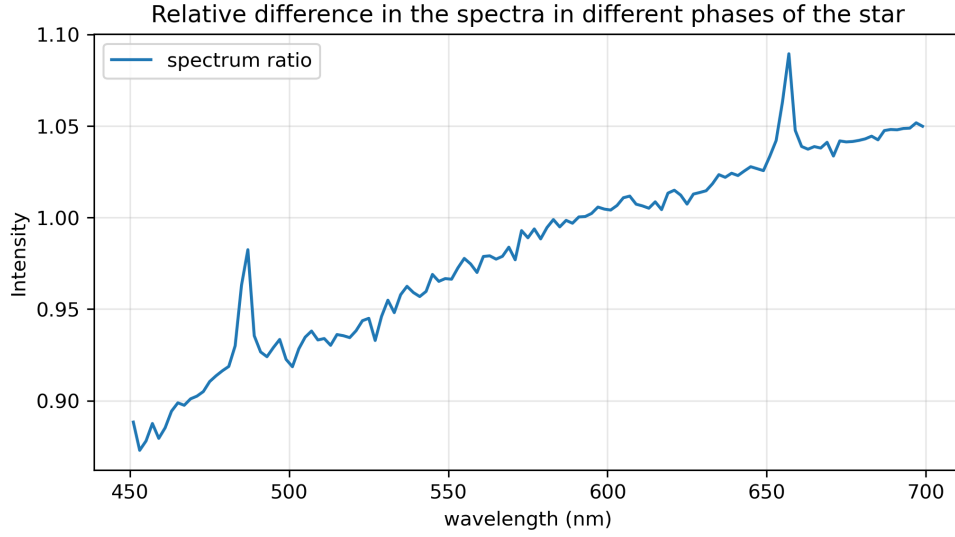


Figure 4.5: Ratio of the spectra in different opposing phases of a star with a surface gravity $\log(g)$ cycling between 2.5 dex and 3 dex and an effective temperature cycling between 6200 K and 6700 K.

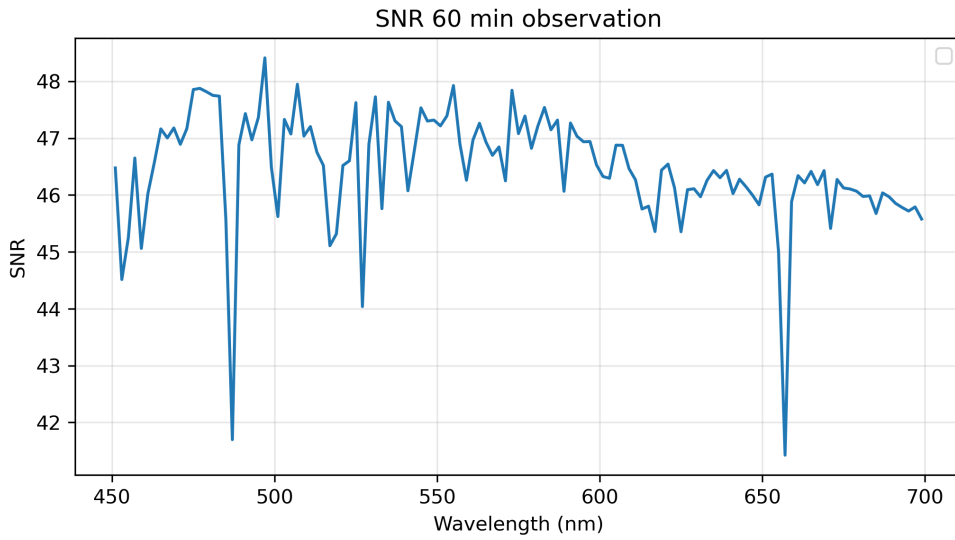


Figure 4.6: SNR of a measurement of a 11 mag star, a background at 16 mag/arcsec² with the proposed background spectra and 60 min exposure time. The noise is only the residual of the subtracted background and dark current, and assumes perfect subtraction.

Chapter 5

Discussion

The proposed setup can be used to analyze spectra of variable stars to determine the phase of a variable star, but is not able to measure the metallicity. What was neglected so far is that the measurement of the period can also be done measuring just the apparent brightness of the star. This is possible without a spectrometer and has been done for more than 100 years[32]. To analyze spectral features in greater detail, a higher resolution spectrometer would be needed, which could be implemented with more elaborate optics. This leads us to an issue with SNR. In our example the difference in the spectra is 5% in the strong spectral features with a signal-to-noise ratio of 2.5%. Figure 5.1 shows the potential fraction of the stellar spectra captured at a spectral resolution of $R\ 400$. Although the peaks are more narrow now, they could have already been resolved before. By increasing the spectral resolution, less pixels can be binned to reduce noise. This leads to an increased SNR of 3%.

To increase the SNR we could also look at brighter targets. This leads to the issue that there are not a lot of regions with numerous possible bright targets in a window of 14 by 9 arcminutes. For targets like the planets, that lay well in the available window, the issue arises of how well we can separate targets. The mirrors on the DMD allow us to select targets of under one arcsecond in size, but on the spectrometer their spot size is increased. The spot sizes shown in the appendix A.5 give an idea of the issue. The separation between different targets must be larger than the spot size on the spectrometer to avoid overlapping spectra, are around 18 arc-seconds along the x axis and just one spectrum can be taken along the y axis. This severely limits the applicable use-cases.

As it becomes apparent that probably no scientific breakthrough will be made using this proposed design, that does not mean all hope for a scientific contribution is lost. The proposed design is simple, can be built for relatively

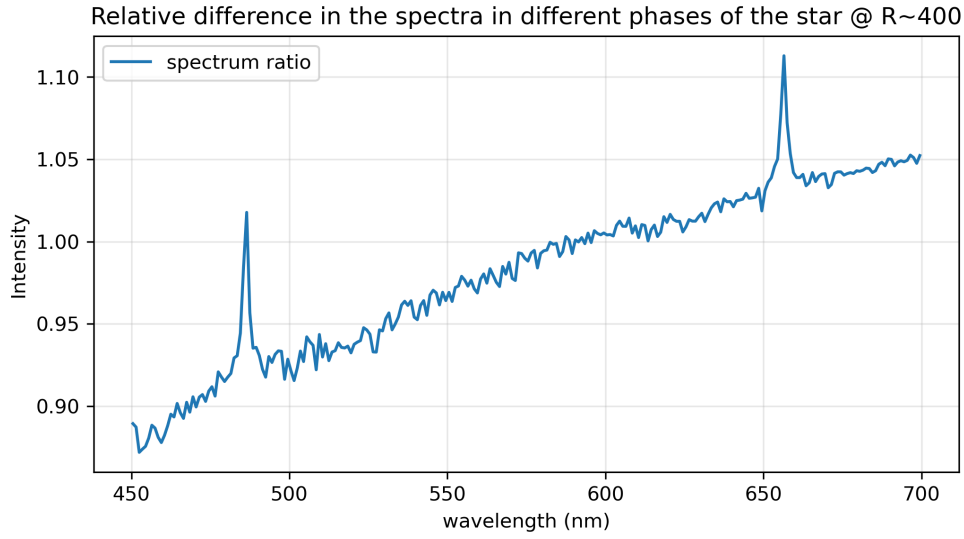


Figure 5.1: Ratio of the spectra in different opposing phases of a star with surface gravity $\log(g)$ cycling between 2.5 dex and 3 dex and an effective temperature cycling between 6200 K and 6700 K. This spectrum is sampled at a resolution of R 400

cheap and can be used with the existing hardware. Due to its restrictions and limitations, it teaches the requirements of a useful science instrument well and can have its place in the education of the people that will make the next actual scientific breakthrough.

As the resolution is high enough to conduct simple experiments, the device could be part of a physics lab, in which the optic design has to be built out, calculations on expected performance are done and compared to the achieved performance in a field experiment.

If this concept is to be turned into a real device substantial work has to be conducted. In a first step the proposed optical setup is to be built out on an optical bench to confirm the simulated results. The response of the device to misalignment and deviating distances between the elements should be evaluated to select suitable materials for the enclosure. The enclosure has to be designed and manufactured up to these requirements. One constraint for the enclosure is that it must be mounted to the back of the existing telescope. This gives constraints for the size and weight of the device. The device is somewhat flexible in this regard, as folding mirrors can be used to redirect light where needed. On the computational side additional work is needed. On the imaging arm the in-camera pixels have to be mapped to the DMD mirrors and a target selection software has to be developed. On the spectrometer side the mirrors of the DMD have to be mapped to the regions where the spectra land. As the spectra are not mapped to straight lines with

equal distances between wavelengths, an elaborate calibration system has to be developed. The spectra have to be calibrated radiometrically, as some surfaces have a wavelength depended throughput.

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List of Figures

- 1.1 The goal of multi-object spectroscopy is to capture spectra of multiple objects in one observation. Most MOS systems follow the same scheme. In a focal plane (1) targets are selected, their light dispersed (2) and the spectra captured (3). 2
- 1.2 Images of the robotically controlled slit plate of the MOSFIRE instrument at the Keck Observatory. The image was sourced from [10]. 3
- 1.3 The Fiber Positioning Unit (FPU) of the Moons instrument allows to precisely position fibers on targets. As the reach of the FPUs overlaps, multiple spectra of locations close by can be obtained. The figure was sourced from the original Moons Paper [1]. . . . 3
- 1.4 Schematics of the integral field spectrometer of the NIRSpec instrument on JWST. The figure was sourced from [18]. 4
- 2.1 The globular cluster Messier 5 observed with a 6 in f/9 amateur telescope. The image was sourced from <https://app.astrobin.com/equipment/explorer/telescope/1097/gso-6-f9-ritchey-chretienon25.08.2024>. 6
- 3.1 Schematics of the micro-mirror array of an DMD. Each micro-mirror can be individually set. The switching between the two stable states can happen at a high frequency. The image was sourced from <https://www.sciencedirect.com/topics/engineering/digital-micromirror-device> 7

3.2	Schematic representation of the opto-mechanic setup using separate optical arms for spectroscopy and imaging. The light from the telescope (1) is split by the DMD (2) into the Spectroscopy setup (left arm) and the imaging setup (right arm). The light in the spectroscopy setup is redirected by a folding mirror (3) onto a collimator lens (4) passes through a grating (5), is refocused by the focusing optics (6) onto the spectroscopy sensor (7). The light in the imaging arm is redirected by a folding mirror (8) and is focused by an imaging lens (9) onto the imaging sensor (10). . . .	8
3.3	Optical layout of the spectroscopic arm. Light from the telescope (1) is modulated by the DMD (2) at the focus of the telescope, redirected via a folding mirror (3) to accommodate the telescope mounting geometry, collimated (4), dispersed by a 600 lines/mm transmission grating (5), focused (6), chromatically corrected by a flat window (7), and detected at the sensor (8). Both lenses have a focal length of 200 mm The optical system was designed using Ansys Zemax OpticStudio. Rays of wavelength 700 nm (red), 580 nm (yellow), 460 nm (blue), from fields with an angle $+0.125^\circ$, 0° , -0.125° were traced.	9
4.1	Spectrum of a 12 mag star simulated with phoenix[26]. The simulated star has an effective temperature of 6200 K, surface gravity $\log(g)=2.5$, $[\text{Fe}/\text{H}]=0$, $[\alpha/\text{Fe}]=0$, corresponding to a RR Lyrae star from [24].	11
4.2	Spectra of different lamps. Samples of street light from a spectral library[28] are used.	12
4.3	Proposed background spectrum at a 16 mag/arcsec ²	12
4.4	Possible retrieved spectrum from the mag 11 star using the proposed device. Two strong absorption features are clearly identifiable. The background and dark images were subtracted and the image normalized so that the highest value is 1.	13
4.5	Ratio of the spectra in different opposing phases of a star with a surface gravity $\log(g)$ cycling between 2.5 dex and 3 dex and an effective temperature cycling between 6200 K and 6700 K.	14
4.6	SNR of a measurement of a 11 mag star, a background at 16 mag/arcsec ² with the proposed background spectra and 60 min exposure time. The noise is only the residual of the subtracted background and dark current, and assumes perfect subtraction.	14
5.1	Ratio of the spectra in different opposing phases of a star with surface gravity $\log(g)$ cycling between 2.5 dex and 3 dex and an effective temperature cycling between 6200 K and 6700 K. This spectrum is sampled at a resolution of R 400	16

A.1	Simulated spot diagram of fields at $+0,125^\circ$, 0° , $-0,125^\circ$, respectively $+0,85^\circ$, 0° , $-0,85^\circ$ at a wavelength of 460 nm.	28
A.2	Simulated spot diagram of fields at $+0,125^\circ$, 0° , $-0,125^\circ$, respectively $+0,85^\circ$, 0° , $-0,85^\circ$ at a wavelength of 570 nm.	29
A.3	Simulated spot diagram of fields at $+0,125^\circ$, 0° , $-0,125^\circ$, respectively $+0,85^\circ$, 0° , $-0,85^\circ$ at a wavelength of 700 nm.	30

Chapter 6

Acknowledgement

This report was written with the help of LLMs. While the text was fully drafted by humans, AI was used for type checking, grammar correction, and removal of incomplete sentences. The code for the calculations was written with a coding agent. AI was used to accelerate the learning curve in Zemax.

I thank Dr. Adrian Glauser for his support and patience throughout this project.

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Appendix A

Appendix

The supplementary materials are made available on Github under <https://github.com/Mdmdma/AmateurMOS>

A.1 Zemax configuration

The Zemax configuration used for the performance estimation is saved as a table. As the table is very large, it is not shown here, but can be found as a csv in the supplementary material in the `optical_setup/zemax_lens_data` folder. Due to a NDA the exact telescope configuration can not be given, but as the telescope resolution is much higher than the spectrometer resolution, the error from using a parallax with the 3454 mm focal length is low enough.

A.2 Code

All code was written in python. As the code is structured in a script file and a helper file it is not given here, but can be found in the supplementary material under `src/stellar_spectra_simulation`.

A.3 DMD datasheet

As no compact datasheet is available, it is not given here. The long form datasheet can be found under `datasheets/dmd_datasheet`.

A.4 Telescope datasheet

The telescope datasheet can be found under `datasheets/telescope_datasheet` in the supplementary materials.

A.5 Simulated spot diagrams



Figure A.1: Simulated spot diagram of fields at $+0,125^\circ$, 0° , $-0,125^\circ$, respectively $+0,85^\circ$, 0° , $-0,85^\circ$ at a wavelength of 460 nm.

A.5. Simulated spot diagrams



Figure A.2: Simulated spot diagram of fields at $+0,125^\circ$, 0° , $-0,125^\circ$, respectively $+0,85^\circ$, 0° , $-0,85^\circ$ at a wavelength of 570 nm.

A.5. Simulated spot diagrams

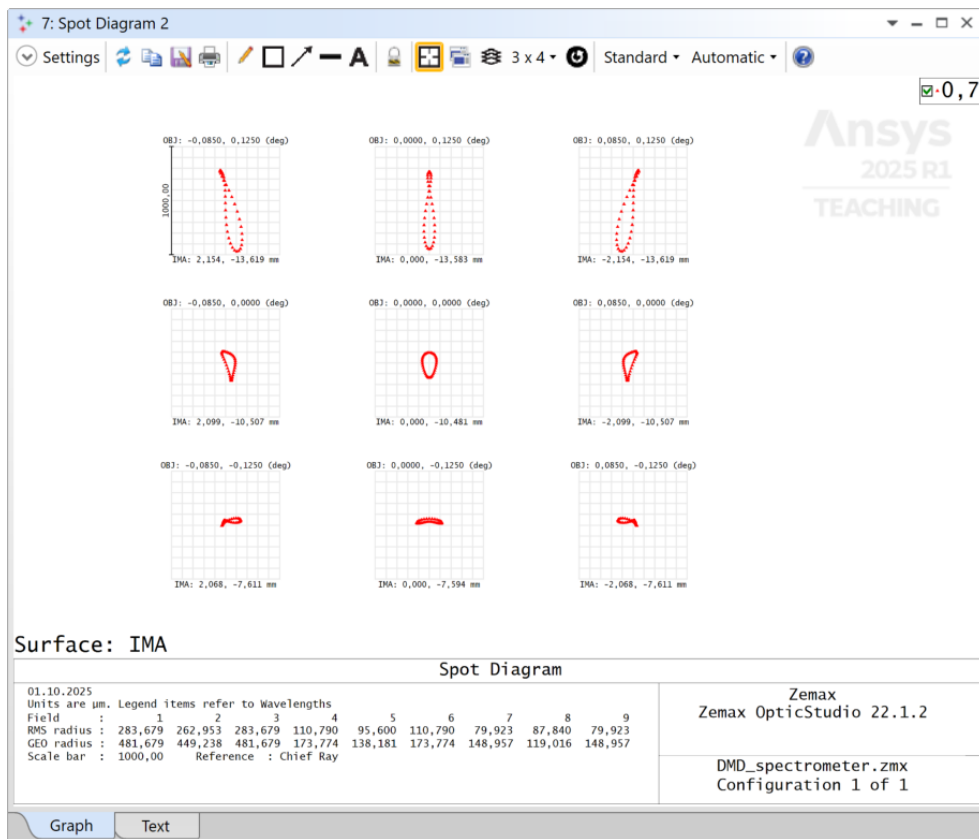


Figure A.3: Simulated spot diagram of fields at $+0,125^\circ$, 0° , $-0,125^\circ$, respectively $+0,85^\circ$, 0° , $-0,85^\circ$ at a wavelength of 700 nm.